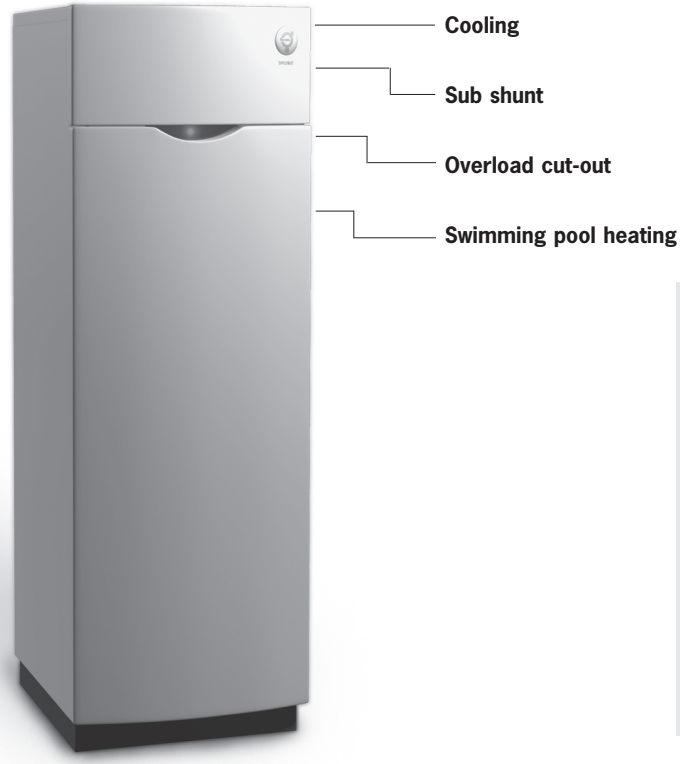
Thermia Diplomat Expansion Card



Thermia Diplomat

Instructions

9680-55173001

Rev 3

Thermia Värme AB reserves the right to make changes in components and specifications without prior notice.

A Thermia Diplomat expansion card installed on your heat pump enables the following added functions.

1.Overload cut-out.

Reduces auxiliary heating to an output corresponding to the preset maximum current.

2. Cooling.

The heat pump can also be used for cooling during the summer; either passive cooling (direct from the borehole) or via a combination whereby the borehole is utilized as long as it remains cool enough, then active cooling (the heat pump's compressor is used to cool the medium, like a refrigerator).

3. Sub shunt.

This is used when the heating system has to control a second temperature level; e.g., when radiators are combined with underfloor heating or when a pool group is used. The temperature in the sub shunt is controlled according to a separate heat curve.

4. Pool

Used to control the temp in a swimmingpool. The temp in the pool is controlled by a separate sensor independet of the heating of the house and hotwater.

5. Sum alarm

For external alarm monitoring. Sends a 230V output signal for every alarm event.

6. 0-10V output voltage for frequency control.

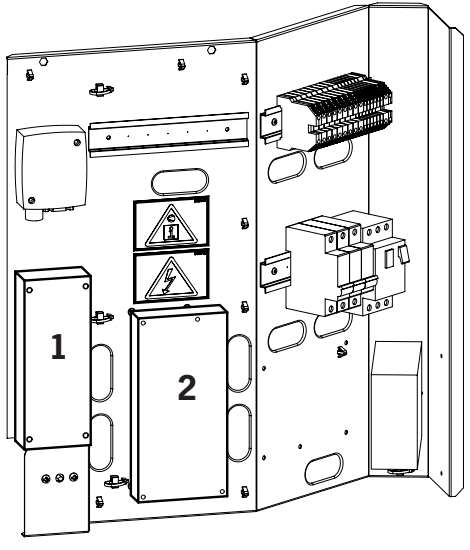
A signal that is controlled by the heating system's energy balance to enable variation of the capacity of the heat pump's compressor.
NOTE! This function is intended for future use, since today's compressors are not designed for frequency-controlled capacity variation.

Connecting the Expansion Card to the Control Unit

The expansion card (1) is placed in the intended slot in the electric compartment and connected to the control unit with the accompanying communication cable. The communication cable is fitted on both ends with connectors that are connected to the connection points marked ext.Com on the control unit and expansion card.

Function/Connection Overload cut-out.

Connect the current sensor to the phases in the fuse group to be protected, according to the circuit diagram. Set MAX AMPS on the Auxiliary Heating menu (see installation instructions for Thermia Diplomat, p. 31). Auxiliary heating is then reduced if current exceeds the preset value for MAX AMPS. The current margin is 5A. Stepping-up is delayed by one minute if current has been within the margin range. If the heat pump is starting up, the current limit is temporarily stepped down by 5A.



Function/Connection Cooling.

PASSIVE COOLING

Connect sensors, shunt motor and circulation pump according to the circuit diagram. Set desired temperature on the cooling system (see installation instructions for Thermia Diplomat, p. 30). The cooling function is set with the help of the heat pump's NODE address (see installation instructions for Thermia Diplomat, p. 33).

Passive cooling without room sensor

NODE: 19 = always passive cooling if outdoor temperature exceeds value for HEATSTOP.

Passive cooling with room sensor

NODE: 2 – 18 = Passive cooling starts if room temperature is 2°C above desired room temperature and stops at 0.9°C above desired room temperature.

ACTIVE/PASSIVE COOLING

Connect sensors, shunt motor, reversing valves and circulation pump according to the circuit diagram. Set desired temperature on the cooling system (see installation instructions for Thermia Diplomat, p. 30). The cooling function is set with the help of the heat pump's NODE address (see installation instructions for Thermia Diplomat, p. 33).

Active/passive cooling without room sensor

NODE: 20 = always passive cooling if outdoor temperature exceeds value for HEATSTOP. Active cooling if BRINE IN temperature exceeds desired preset COOLING temperature for 2 minutes.

Active/passive cooling with room sensor

NODE: 21 – 30 = Passive cooling starts if room temperature is 2°C above desired room temperature and stops at 0.9°C above desired room temperature. Active cooling starts if room temperature is 4°C above desired room temperature and stops at 1.9°C above desired room temperature.

Function/Connection Sub shunt.

Connect sensors and shunt motor according to the circuit diagram. The new sub shunt's heat curve is set via the heat pump's Heat Curve menu (see installation instructions for Thermia Diplomat, p. 29). Under the Heat Curve menu is Heat Curve2, where the following parameters can be set for the sub shunt: CURVE2 = desired sub shunt temperature at an outdoor temperature of 0°C; MIN = lowest temperature for the sub shunt; MAX = highest temperature for the sub shunt. If the sub shunt is to be used to maintain a constant temperature, e.g., for a pool group, both MIN and MAX are set to the desired value. Bending of curve and heatstop are the same as CURVE.

Function/Connection Sum Alarm.

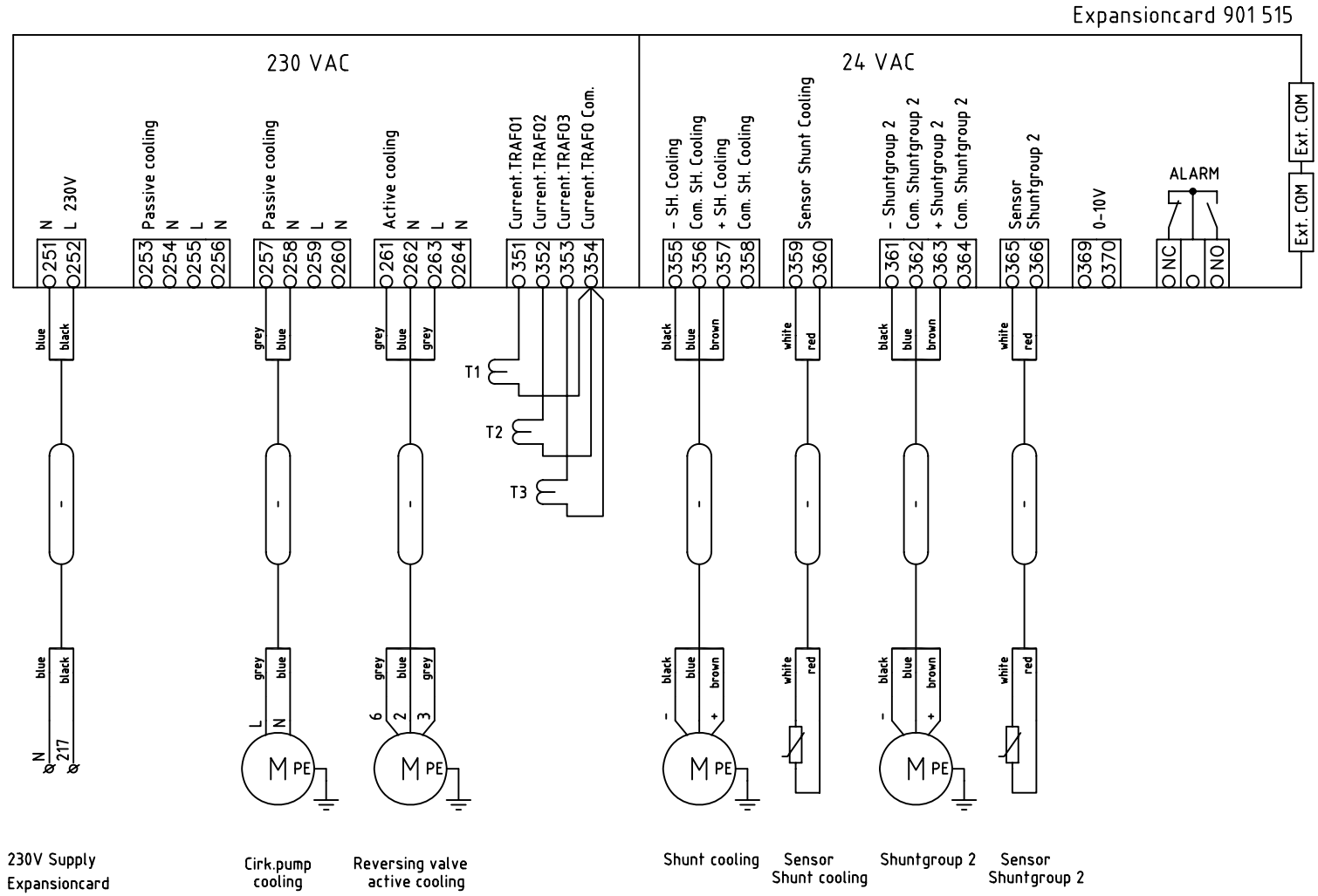
Connect the external device to the output port for sum alarm according to the circuit diagram. When any of the possible alarm events occurs, a 230V (max 5A) output signal is emitted.

Function/Connection Frequency Control 0-10V.

This function is not currently used, since today's compressors are not designed for frequency-controlled capacity variation.

Function/Connection pool

The NODE adress of the heat pump must be set to 11 to activate the pool function.
Connect the sensor and 3-way valve according to the electrical schedule. The pool temt is set in the controller of the heatpump under the menu HEAT CURVE (see the installation manual). In the menu HEAT CURVE is a setting named pool where the desired pool temperature can be set. If the value is set lower than 5°C a ⏻ symbol is shown indicating the pool function is shout off. In this menu is also "Pool hystereses" located, this value is the gap between on/off for the pool sensor. Control that there always is a flow circulating over the pool sensor while in pool mode.



Thermia

Wiring diagram
Expansioncard 901 515

External connections

DRAWING NO. 54201 EN		
CONSTRUCTED BY Jael	CREATED BY Jael	SHEET 1
DATE 2005-01-25		NEXT SHEET -

If these instructions are not followed during installation, operation and maintenance, the obligations of Thermia Värme AB under the warranty regulations are no longer binding.

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